

A Molecular Dynamics Investigation of Wax Crystallization in Crude Oil under Varying Pressures, Temperatures, and CO₂ Contents

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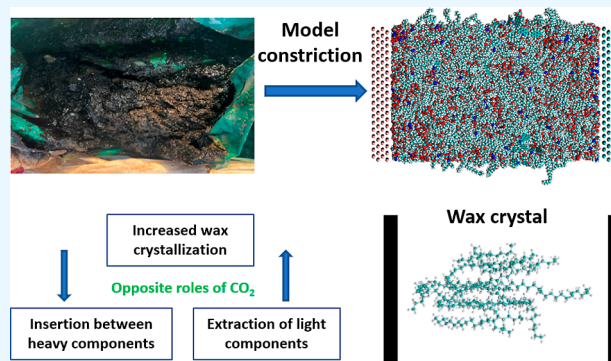
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ABSTRACT: Wax precipitation in crude oil poses a significant flow assurance challenge, leading to a reduced level of production and operational blockages. This study employs molecular dynamics (MD) simulations to investigate the molecular mechanisms of wax crystallization under varying wellbore conditions. A multicomponent crude oil model was constructed based on samples from the Mahu oilfield, and simulations were performed at different depths (0 to 3400 m) and CO₂ concentrations (0% to 50%). The results reveal that wax crystallization, quantified by a decrease in the diffusion coefficient and an increase in cluster size, intensifies as the pressure and temperature decrease with wellbore depth. A complex, dual role of CO₂ was identified, where it not only acts as a light component at high pressures, inhibiting wax formation, but also extracts light components from the oil, promoting aggregation of heavier molecules. This results in the maximum crystallization observed at 30% CO₂ under flash separation conditions. To reconcile discrepancies with field observations, a differential separation (“degas”) route was simulated, which demonstrated significantly enhanced crystallization, particularly at 50% CO₂. These findings underscore the importance of the thermodynamic pathway in predicting wax deposition and suggest that maintaining high pressure and temperature and managing the CO₂ concentration are key strategies for mitigation.



INTRODUCTION

Wax precipitation in petroleum is a critical flow assurance challenge that significantly impacts the oil and gas industry. This phenomenon occurs when high-molecular-weight paraffins, primarily *n*-alkanes, crystallize and deposit from crude oil as it cools below a certain threshold.¹ These solid deposits can adhere to pipeline walls and other production equipment, leading to reduced flow rates, increased pumping costs, and, in severe cases, complete pipeline blockages that result in substantial operational and economic losses.^{2,3} The formation of wax deposits not only narrows the effective pipe diameter but also alters the rheological properties of the oil, increasing its viscosity and promoting gel formation, which further impedes flow.⁴

Several key factors influence the propensity for wax precipitation, with crude oil composition, temperature, and pressure being the most significant.^{5,6} The concentration of long-chain *n*-alkanes (typically C₁₄₊) has been shown to correlate linearly with the WAT.¹ The solubility of wax in the oil matrix is also composition-dependent, with lighter oils generally dissolving wax more effectively than heavier oils.^{7,8} In terms of temperature, the precipitation process is governed by thermodynamic and kinetic factors. Thermodynamically, wax formation is driven by the reduced solubility of paraffins as temperature decreases.^{9,10} The onset of wax-related operational

issues is marked by the wax appearance temperature (WAT), defined as the temperature at which the first wax crystals become visible upon cooling.¹¹ Understanding the complex interplay between WAT and various factors is, therefore, essential for designing effective mitigation strategies. Kinetically, factors such as the cooling rate and shear history modulate the crystallization process, affecting the size, morphology, and deposition behavior of the wax crystals.^{12,13}

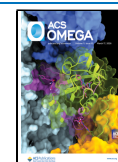
While the temperature is the primary driver, pressure also plays a crucial role. Generally, an increase in pressure reduces the solubility of wax in crude oil, leading to a higher WAT.^{14,15} This relationship is often nonlinear and is further complicated by the presence of dissolved gases, such as methane or carbon dioxide, which can alter the phase behavior of the oil under pressure.¹⁶ Moreover, the presence of other crude oil constituents, such as asphaltenes and resins, adds another layer of complexity.^{17,18} Asphaltenes can coprecipitate or adsorb onto wax crystal

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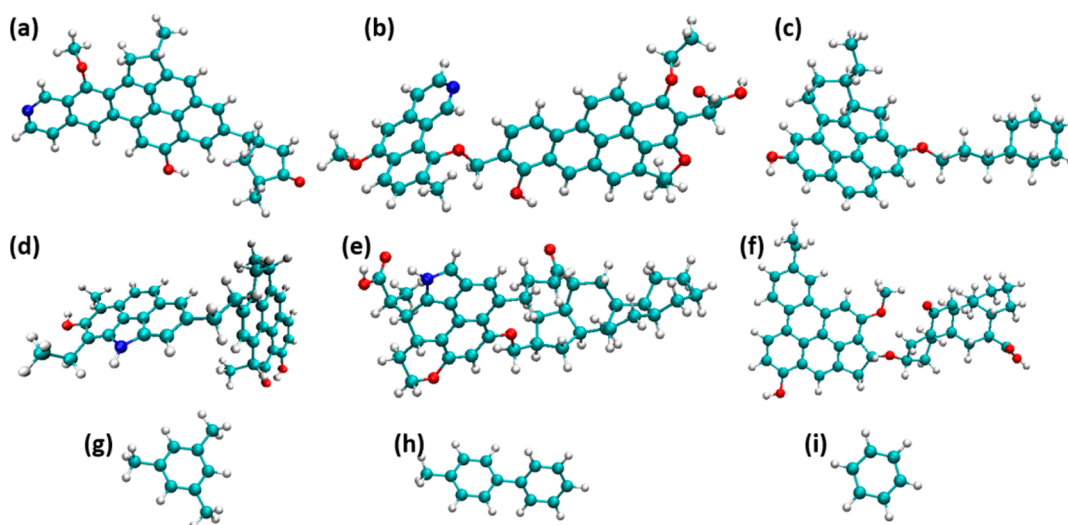


Figure 1. Model molecules used to construct the oil phase, where (a–f) represent the asphaltene and resin components, and (g–i) represent the aromatics.

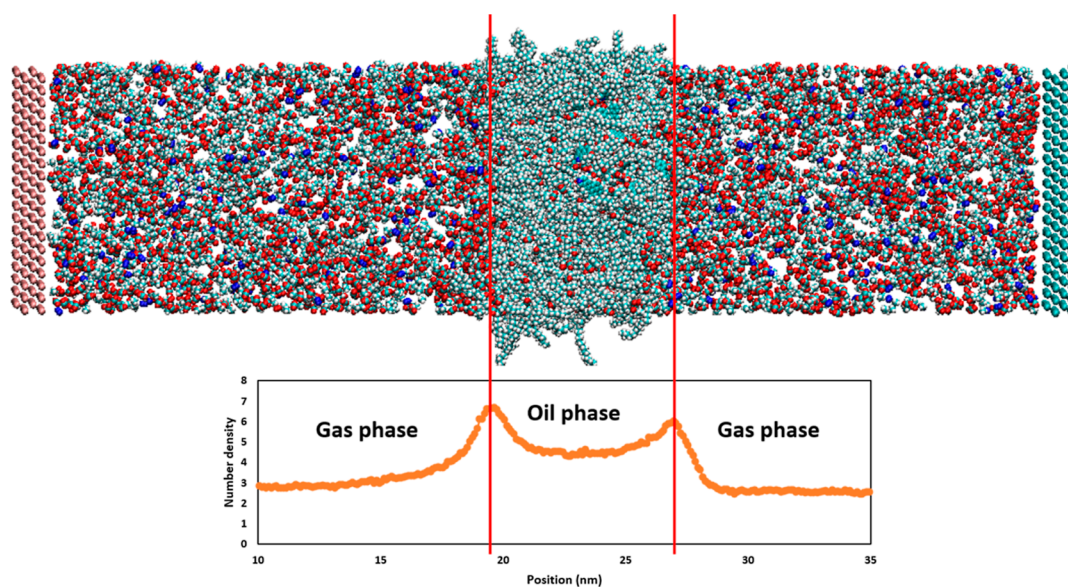


Figure 2. Configuration of the model system, where two pistons were positioned at both ends of the oil and gas mixture. At given temperature and pressure conditions, the mixture exhibit as a two-phase system. The oil phase stands in the middle, while the gas phases are distributed on both sides of the oil phase. The oil–gas boundaries can be determined by the peak position of CH₄ density along the simulation box.

surfaces, either inhibiting or exacerbating deposition depending on their concentration and molecular structure.^{4,19,20}

Wax precipitation was also computationally investigated. Thermodynamic methods typically treat wax precipitation as a bulk phase equilibrium, without explicitly accounting for molecular interactions driving nucleation, growth, and aggregation. In contrast, molecular dynamics (MD) simulations have emerged as a powerful tool for providing atomic-level insights into wax aggregation and deposition under various thermodynamic and flow conditions.^{21–24} MD simulations reveal dynamic interactions among CO₂, light hydrocarbons, and wax molecules beyond the scope of thermodynamic models. Furthermore, MD allows investigating how asphaltene–resin aggregates may inhibit or promote wax crystal growth via steric effects and surface adsorption. These mechanisms remain difficult to parametrize in thermodynamic models. MD simulations can explicitly model the influence of *n*-alkane chain length, crude oil

composition, and the presence of inhibitors or other components like asphaltenes on crystallization behavior.^{25,26}

For example, Esselink et al.²¹ studied the crystallization and melting temperatures of alkanes ranging from butane to dodecane and found that the chain length of *n*-alkanes significantly influences their aggregation patterns. Their work highlighted the size dependence of crystallization rates, with longer chains exhibiting more pronounced nucleation behavior. Similarly, Waheed et al.²⁷ observed phase transitions in *n*-eicosane, where amorphous regions transformed into a close-packed hexagonal structure in the presence of a crystal surface, underscoring the role of temperature in crystallization kinetics. These findings align with that by Zerze et al.,²⁸ who identified metastable nuclei in eicosane systems at 250 K, emphasizing the quasi-equilibrium states that precede abrupt crystallization. Additionally, Takeuchi²⁹ studied the induction period of crystallization in short-chain alkanes, revealing that local parallel

order developed before macroscopic changes became apparent. The MD approach was also applied to understand, design, and optimize wax inhibitors. For instance, Vijayakumar and Ridzuan²⁶ utilized MD to analyze the intermolecular interactions between *n*-eicosane and surfactant–nanoparticle blends, revealing how hydrogen bonding and van der Waals forces dictate wax solubility. Together, these studies demonstrate how MD simulations can unravel the molecular-scale dynamics of wax aggregation, offering a foundation for predicting and mitigating deposition in industrial applications.

The rest of the article is structured as follows. First, the compositions of crude oil and natural gas are analyzed, and a molecular model is built accordingly. Simulation setups are introduced, and the results are rationalized to elucidate the influence of depth and CO₂ concentration. The difference between MD result and field observations is discussed and validated by additional simulations. Conclusions and recommendations are then given.

SIMULATION METHODS

Methods

Composition of the Model Systems. The simulated fluid was constructed as follows. The crude oil and natural gas were sampled from a well in the Mahu Conglomerate Oilfield in Xinjiang, China, and gas chromatography (GC) analyses were conducted to obtain the compositions. The gas oil ratio (GOR) was also acquired as the average GOR of the past 30 days before the date of sampling. Molecular models were built based on each of the detected components for the gas and oil phases. The number of molecules for each component was determined based on the average GOR, and the molecules were added into a simulation box. Saturates, aromatics, resin, and asphaltene (SARA) analysis was conducted to obtain the ratios of aromatics, resins, and asphaltenes in the crude oil sample, and corresponding molecules were added to the model mixture. Components used to represent the aromatic, resin, and asphaltene moieties are illustrated in Figure 1.

It is acknowledged that these are not faithful representations of the oil sample; however, such compromises must be made due to the challenges in determining resin and asphaltene structures. The concentration of CO₂ within the context of this work is defined as the molar composition of CO₂ in the gas phase before mixing with the oil phase. Model fluid systems with five CO₂ concentrations, from 10% to 50% at 10% intervals, were constructed.

Simulation Setups. Instead of directly using the *NPT* ensemble, an *NVT* ensemble coupled with two pistons^{30–32} were utilized to achieve different pressures with the purpose to minimize the equilibration time that is usually longer for *NPT* simulations. As shown in Figure 2, the left piston remains fixed throughout the simulations, while the right piston was given leftward accelerations, corresponding to different pressures, to compress the fluids in between. Although this approach breaks the periodic boundary along the axial direction, which should not theoretically impact the phase behavior, it allows for almost instant establishment of given pressures. A comparison between *NVT* and *NPT* methods at 30% CO₂ concentration is given in Table 1, and the result shows negligible deviation. Therefore, the *NVT*-piston approach was

adopted for all simulations in this work. The CHARMM force field³³ parameters were chosen for all organic molecules, for its outstanding agreement with experimental shear viscosity and diffusion coefficients. GROMACS³⁴ software (2024.3) was used to conduct all simulations.

Five sets of pressure and temperature combinations, corresponding to the reservoir condition (3400 m), 2000 m, 1000 m, 300 m, and wellhead conditions (0 m), were simulated for systems containing 10%, 20%, 30%, 40%, and 50% CO₂, respectively. The temperature and pressure conditions are given in Table 2. Each simulation was run for 50

Table 2. Temperature and Pressure Conditions at Different Wellbore Depths

wellbore depth (m)	pressure (bar)	temperature (K)
0	51.9	282.0
300	62.2	289.1
1000	88.8	305.7
2000	151.9	332.3
3400	448.0	357.2

ns and consecutively followed the order of formation condition, 2000 m, 1000 m, 300 m, and wellhead conditions. To be specific, simulations at the reservoir condition were run for 50 ns, and the resulting system, which is already equilibrated at that condition, was used as the starting point for the 50 ns simulation at 2000 m temperature and pressure conditions. The resulting system was then used as the starting point for the simulation at 1000 m conditions. This sequential arrangement of simulations ensures that each simulation is commenced with an equilibrated state and the starting condition is not drastically different from the target condition to avoid unnatural changes in phase behavior. The gaseous phase is not removed at the end of simulations, and the gaseous and liquid phases are always in contact, making them equivalent to flash separation processes. At the end of each simulation, the mean square displacement (MSD) is calculated for the wax composition (from C₁₈ to C₄₃) and asphaltene and resin compositions, from which the diffusion coefficient can be automatically calculated. The extent of molecular alignment is also analyzed, as will be detailed later in this section.

In the course of this work, simulations under an additional scenario were conducted as follows. The simulations again start with a 50 ns run at reservoir conditions, any gaseous phase is removed at the end of the simulations, and the remaining liquid phase is used as the starting system for temperature and pressure conditions at 2000 m as well as the shallower positions above it. The boundary between gas and oil is considered to be the position where the CH₄ density exhibits a sharp change on the number density curve obtained using the built-in number density function in GROMACS, as demonstrated in Figure 2. This process goes on until the conditions reach that of the well head. This is equivalent to the differential separation of gas from oil.

Determination of Crystallization

The determination of onset of crystallization and the extent of crystallization is realized using a self-developed algorithm. To put briefly, molecules that contain more than 18 carbon atoms (the threshold to be considered part of wax) are analyzed for their segmental linearity; in other words, even if a molecule is in a bent configuration, as long as a segment of it is linear, this segment is recorded as a reference. All linear segments within the simulation box at the end of each run are thus identified and recorded. The segments are then analyzed for clustering. That is, if two segments are within 1 nm of each other, and the cosine of the vector of one segment to that of the other is larger than 0.95, these two segments are considered as being parallel and are considered to be within the same cluster. This searching process is repeated until no more segments are found that simultaneously satisfy the distance (<1 nm) and angle (cos > 0.95) requirements. In this way, the number of clusters within the simulation box and the number of segments, which is usually equivalent to the number of molecules, within the same cluster can be determined. Given that wax crystallization predominantly exhibits as the parallel arrangement of

Table 1. Comparison between *NPT* and *NVT* Methods

depth (m)	gas to oil volume ratio			diffusion coefficient 1×10^{-5} cm ² /s		
	<i>NPT</i>	<i>NVT</i>	error/%	<i>NPT</i>	<i>NVT</i>	error
0	3.6252	3.6483	0.64	0.2008	0.1959	2.44
300	3.1169	3.1513	1.10	0.2958	0.3007	1.66
1000	2.0625	2.0049	2.79	0.5124	0.5092	0.63
2000	1.0011	1.0005	0.06	0.9005	0.9101	1.07
3400	0	0	0	1.8551	1.8531	0.11

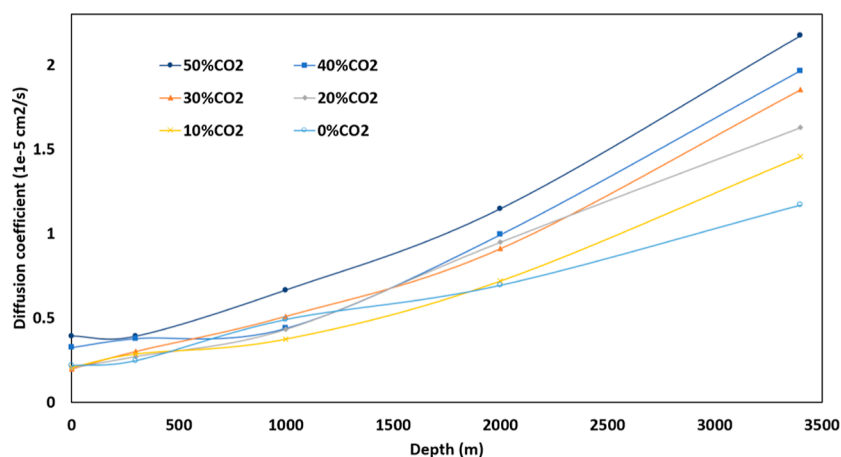


Figure 3. Diffusion coefficients of wax under different wellbore depth and CO₂ concentration conditions.

long alkane molecules, results obtained using the algorithm can be used to evaluate the crystallization of the simulated model system. Shifts in the number of clusters containing different numbers of segments (molecules) is thus a clear indication of the progression of wax crystallization in response to changes in the pressure and temperature conditions, as will be detailed in the [Results and Discussion](#) section.

RESULTS AND DISCUSSION

Influence of Wellbore Depth on Wax Crystallization

A larger diffusion coefficient is indicative of faster movement of molecules, which indirectly corresponds to less wax crystallization. On the contrary, a lower diffusion coefficient suggests the formation of wax clusters. Diffusion coefficient values obtained using the last 20 ns of each 50 runs are compared in [Figure 3](#). A clear trend is that with the decrease in temperature and pressure, the diffusion coefficients of wax (any molecule higher than C₁₇) and asphaltene components show a decreasing trend. Specifically, at reservoir conditions (3400 m), all light components are dissolved in the liquid phase, including CO₂ molecules, if present. The light components are distributed between longer molecules and inhibit cluster formation, resulting in the highest diffusion coefficient. With the decrease in pressure and temperature, the percentage of lighter components in the oil decreased as the gaseous phase began to separate from the liquid phase. As a result, distances between heavier components shrank, facilitating cluster formation.

To elucidate the molecular aggregation process and the spatial relationship between wax components and CO₂ molecules, radial distribution function (RDF) analysis was performed to characterize the local structural organization of CO₂ around wax moieties for each depth corresponding to distinct temperature and pressure conditions. The results shown in [Figure 4](#) demonstrate a progressive increase in the CO₂ density surrounding wax molecules with increasing depth. This trend reflects the improved solvation of CO₂ between wax molecules. Conversely, at shallower depths where temperature and pressure are reduced, CO₂ clustering occurs, leading to microphase separation that diminishes effective contact area between individual wax crystals and dissolved CO₂. Consequently, the RDF curves exhibit only a single, relatively low-intensity peak, indicating weak and nonspecific interactions between CO₂ and the wax components.

For a clear and quantitative comparison of the number of clusters and the number of molecules within clusters, all simulation runs were analyzed, and the data are given in [Figure 5](#).

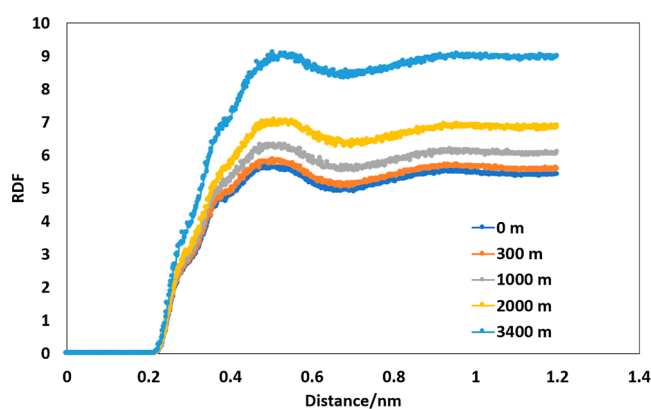


Figure 4. Radial distribution function of CO₂ molecules around wax components at different well depths, showing more prominent CO₂ distribution at deeper wellbore conditions.

A general trend is that bigger clusters are formed at shallower well depth, giving the distribution of the data a triangular shape in the table. In other words, with the reduction in well depth, smaller clusters predominantly containing two or three molecules appeared first and gradually shifted toward higher molecule numbers. During this process, the clusters grew in size with decreasing mobility. This general trend falls in line with the wax precipitation observed on site with decreasing depth. However, in the presence of 50% CO₂, the number of clusters with sizes beyond 3 molecules was minimal, and no apparent crystallization was observed. On the contrary, in the presence of 30% CO₂, a clear wax aggregation was formed that contained 187 molecules in a single cluster demonstrated in [Figure 6](#). Crystallization was also observed for the system with 20% CO₂, which contains a cluster with 51 molecules at wellhead conditions. For systems with 10% CO₂ and no CO₂, crystallization disappeared and the biggest cluster contained only 10 and 7 molecules, respectively. This is contradictory with field observations, where wells that produce gases with a high CO₂ content exhibit more severe waxing issues.

Influence of CO₂ Concentration on Wax Crystallization

To more directly demonstrate the influence of CO₂ on crystallization, the same data in [Figure 3](#) is replotted against CO₂ concentration in [Figure 7](#). It becomes clear that, at formation conditions (3400 m), the wax diffusion coefficient increases with the CO₂ concentration, which is against field

0% CO ₂		No. of clusters at					
No. of molecules in a cluster		0m	300m	1000m	2000m	3400m	
		85	26	33	33	23	
2		6	8	7	3	6	
3		2	5	3			
4		1			1		
5		1					
6		1					
7		1					

10% CO ₂		No. of clusters at					
No. of molecules in a cluster		0m	300m	1000m	2000m	3400m	
		37	34	24	27	30	
2		9	7	7	4	1	
3			1	2		1	
4		1					
5				1			
6							
10			1				

20% CO ₂		No. of clusters at					
No. of molecules in a cluster		0m	300m	1000m	2000m	3400m	
		29	27	26	30	22	
2		15	8	2	3	6	
3		1	1	5	2		
4				1			
5					1		
6			1				
19							
51		1					

30% CO ₂		No. of clusters at					
No. of molecules in a cluster		0m	300m	1000m	2000m	3400m	
		21	36	40	31	23	
2			8	3	4	6	
3				2	1	1	
4							
187		1					

40% CO ₂		No. of clusters at					
No. of molecules in a cluster		0m	300m	1000m	2000m	3400m	
		32	26	22	35	25	
2		6	9	6	2	2	
3		1	2				
4		1					
5		1					

50% CO ₂		No. of clusters at					
No. of molecules in a cluster		0m	300m	1000m	2000m	3400m	
		35	33	30	26	26	
2		15	4	3	1	2	
3		1	1				
4		1					
7		1					

Figure 5. Summary of cluster numbers and molecule numbers within clusters for all simulations. Triangular shapes marked with dotted red lines are to demonstrate the trend of cluster formation.

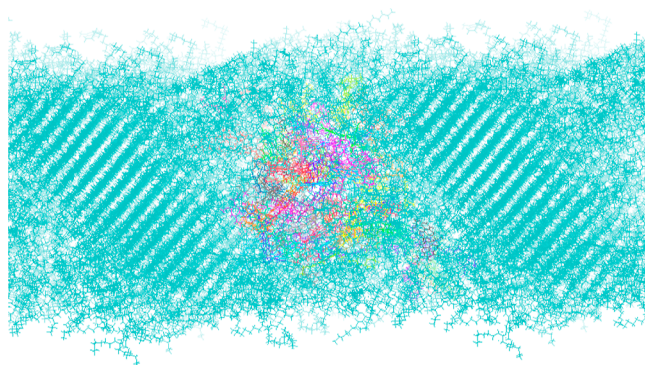


Figure 6. Snapshot of the simulation box containing a wax crystal with 187 alkane molecules. The simulation box was purposefully presented with periodicity along the axial direction to include the whole crystal. Alkane molecules are presented in green, while all other colors indicate aromatic, asphaltene, and resin compositions.

observations. A possible explanation is that the solubility of CO₂ in oil is sufficiently high, and increasing pressures press a large amount of CO₂ molecules into the oil phase. The CO₂ molecules then function as light components in the same way as CH₄ and insert between the heavier components, inhibiting wax formation. In comparison, at wellhead conditions, there is a slight decrease in the wax diffusion coefficient from 0% to 30% CO₂ concentration, followed by an increase at higher CO₂ concentrations. All other depths essentially exhibit a trend in between these two extreme conditions. One possible reason is that although CO₂ “dilutes” the heavier components, it also tends to extract light components from the oil phase into the gas phase, rendering the oil phase heavier. This is exhibited in the interaction energies among wax, light components, and CO₂

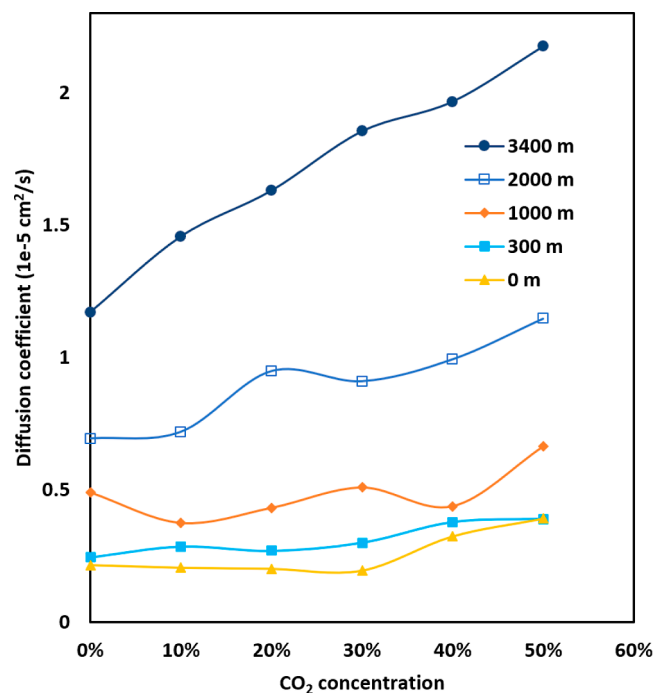


Figure 7. Diffusion coefficients of wax under different wellbore depth and CO₂ concentration conditions, plotted against CO₂ concentration.

molecules shown in Table 3. Under all concentrations, the interaction is stronger between CO₂ and light components, which explains the physical basis for CO₂'s ability to extract light components.

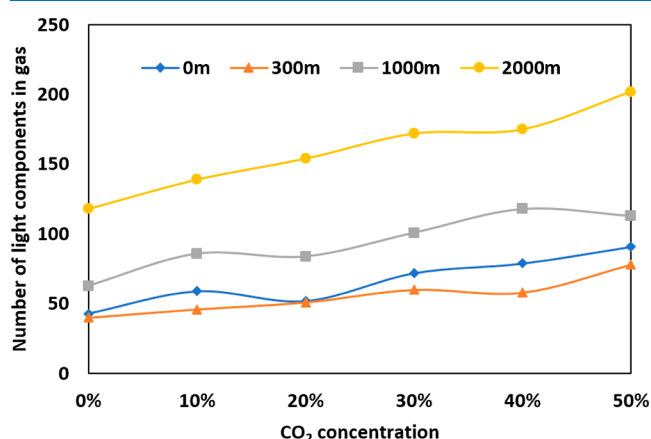
These two opposite mechanisms compromise each other, making 30% of the CO₂ concentration a threshold. Below 30%

Table 3. Interaction Energies, Calculated as the Sum of van der Waals and Electrostatic Interactions, between Wax and CO₂, as Well as Light Components and CO₂, for Systems Containing (a) 10%, (b) 30%, and (c) 50% CO₂

depth (m)	interaction energy at 10% CO ₂ (kJ/mol)		interaction energy at 30% CO ₂ (kJ/mol)		interaction energy at 50% CO ₂ (kJ/mol)	
	wax–CO ₂	light components–CO ₂	wax–CO ₂	light components–CO ₂	wax–CO ₂	light components–CO ₂
0	−761.525	−3018.13	−2364.56	−9,858.51	−3671.03	−16,101.1
300	−791.946	−3162.75	−2708.16	−10,632.9	−4630.46	−17,873.4
1000	−836.275	−3500.48	−2862.37	−11,709.2	−4898.61	−19,583.7
2000	−927.098	−4123.33	−3189.9	−13,991.5	−5487.82	−23,738.3
3400	−1177.32	−5223.62	−4068.79	−17,904.6	−7030.82	−30,685.1

CO₂ concentration, extraction of light components dominates, while beyond 30% CO₂ concentration, CO₂ dissolution overbalances the extraction mechanism.

To validate the above hypothesis, the number of light molecules (C₅–C₁₆) in the gaseous phase was analyzed for all simulations, and the results are presented in Figure 8. It is clear

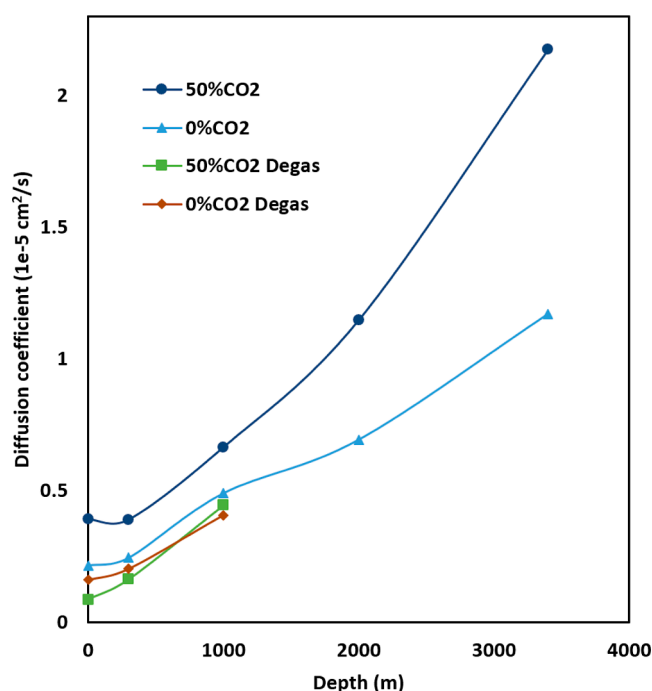
**Figure 8.** Number of light components within the gas phase for different wellbore depths and CO₂ concentrations.

that at all depth conditions, the number of light components in the gaseous phase increases with increasing CO₂ concentration, confirming that the presence of CO₂ extracts light components into the gaseous phase.

It should be mentioned that the above simulations resemble the flash separation of oil and gas where the two phases are always in contact with each other. Considering the usually faster gas flow compared to oil within wellbores, fluid flow within wellbores can be considered as a combination between flash separation and differential separation, with the latter meaning that the gas phase is constantly removed the moment any gas is separated from the oil phase with decreasing pressure. In two-phase flow within wellbores, various flow patterns may occur, including bubbly, slug, churn, and annular flow.³⁵ Regardless of the flow regime, the gas phase consistently rises at a faster rate than the oil due to its lower density and higher mobility, resulting in the progressive degassing of the oil as it ascends toward the surface. Consequently, the oil becomes increasingly degassed with decreasing depth, especially near the top of the wellbore. Given these dynamics, it is reasonable to model the phase separation process as resembling flash vaporization near the bottomhole, where initial pressure reduction causes partial gas exsolution, while the upper sections of the wellbore should experience more of a differential separation due to continuous gas release along the flow path.

An additional set of simulations were conducted for 0% and 50% CO₂ concentrations, where the gas phase was removed at

the end of each simulation before the pressure and temperature conditions were adjusted to those at shallower positions. This process will be addressed as degas hereafter. The results are compared in Figure 9. No gas phase exists under reservoir

**Figure 9.** Comparison between flash separation and degas separation, showing that the diffusion coefficients of wax are significantly lower than those at flash separation conditions.

conditions, and the gas phase was minimal for 200 m conditions; therefore, the degas curve was only applicable between 0 and 1000 m conditions. This results in significantly lowered wax diffusion coefficients, especially that at 50% CO₂ concentration, where substantial crystallization was also observed, which coincides with the observation on site. The dual effect of CO₂ arises from its contrasting behaviors under different thermodynamic conditions. Under high-pressure conditions typical of reservoir environments, CO₂ acts as a wax inhibitor by insertion between long-chain alkanes and delaying wax crystallization onset. This is evidenced by the higher diffusion coefficients of wax observed in high-pressure simulations compared to degassing scenarios. However, upon pressure reduction during production operations, CO₂ undergoes phase separation and selectively extracts the lighter hydrocarbon components from the oil phase. This compositional shift increases the effective concentration of wax precursors in the remaining oil, accelerating wax molecule aggregation and cluster formation. Thus, while CO₂ initially retards wax crystallization, its

subsequent release creates conditions favorable for enhanced wax precipitation. This finding is in agreement with the experimental and computational evidence that emphasizes the solubility-dependent impact of CO₂ on wax appearance temperature (WAT)³⁶ and that thermodynamic pathways significantly influence deposition outcomes.³⁷

Implications and Research Suggestions

To evaluate the influence of asphaltene and resin components on wax crystallization dynamics, molecular simulations were conducted under conditions both with and without these polar fractions. The results shown in Figure 11 indicate that the

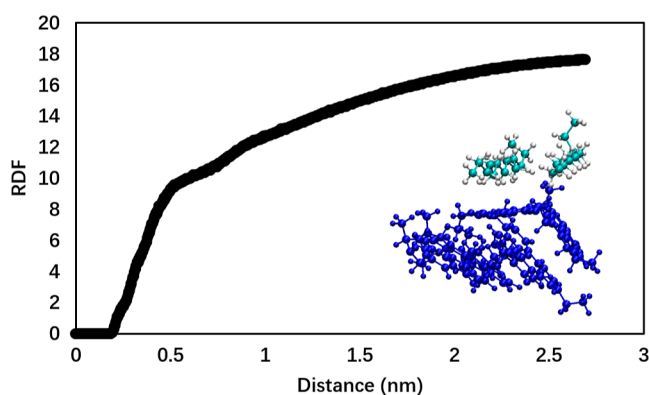


Figure 10. RDF curve of wax around asphaltene in the presence of CO₂ at 30% and 300 m depth conditions. The inset shows a snapshot of two wax molecules (green and white) adsorbed on an asphaltene cluster (blue).

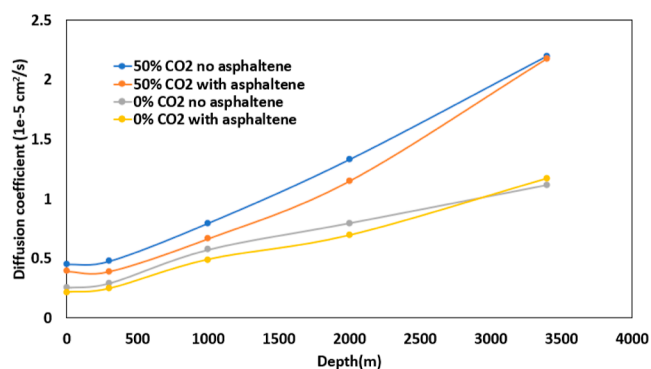


Figure 11. Diffusion coefficients of CO₂ at 0% and 50% concentrations and different depth conditions.

diffusion coefficient of wax molecules is marginally lower in systems in the absence of asphaltenes, suggesting that the presence of asphaltenes promotes wax nucleation and aggregation. A snapshot showing the affinity of a wax cluster for the asphaltene cluster is presented in Figure 10. However, no clear peak can be observed in terms of RDF analysis, indicating interactions between the asphaltene and hydrocarbons. This is understandable in that the asphaltene and resin molecules in this work are predominantly aromatic and polar with a relatively small percentage of aliphatic moieties. It is possible that the interactions between asphaltene and wax can exhibit a different trend if more aliphatic branches are added to the asphaltene and resins.

However, despite this observed trend, it is noted that all asphaltene and resin molecules used in this study underwent rapid self-aggregation, forming a single dominant cluster (Figure

6). As a result, no significant spatial partitioning or physical barrier effect was possible to impede wax growth as suggested in the literature.¹⁸ Consequently, the hypothesized steric hindrance mechanism could not be validated within the scope of this work. Therefore, while preliminary evidence points to a facilitative role of asphaltenes in wax precipitation, the specific molecular interactions, conformational arrangements, and potential for structure-directing behavior between asphaltenes, resins, and *n*-alkane waxes merit further and more systematic investigation.

In terms of practical implications for field productions, based on the results generated in this work, it can be inferred that since wax mainly precipitates at shallower depths, priority should be given to implementing wax control measures in the upper sections of the wellbore. In addition, the pressure distribution along the wellbore should be optimized. Considering that differential separation is a key factor leading to changes in fluid properties and intensified wax deposition, it is recommended that the wellhead backpressure be appropriately increased within operational limits. This helps to delay the upward shift of gas–liquid separation. By maintaining CO₂ dissolved in the oil phase, its solvent effect can be utilized to suppress deep wax precipitation and reduce premature release of light hydrocarbon components. Also, although a peak precipitation tendency was observed at 30% CO₂, in practice, it should be mentioned that the 30% threshold CO₂ concentration is identified as a result of specific oil and gas compositions and *T* and *P* conditions, is influenced by several factors, and should not be generalized as a universal conclusion. The threshold CO₂ concentration needs to be adjusted dynamically according to the specific crude oil composition and separation paths in field applications.

CONCLUSIONS

In this study, molecular dynamics simulations were conducted to provide an atomic-level understanding of wax crystallization in crude oil under various wellbore temperature, pressure, and CO₂ concentration conditions. The key conclusions are as follows:

- As pressure and temperature decrease from reservoir to wellhead conditions, the mobility of wax molecules (C₁₈₊) decreases, as indicated by lower diffusion coefficients, leading to the formation of larger and more crystalline clusters. This finding aligns with field observations of increased wax deposition in shallower parts of the production system.
- CO₂ exhibits a complex, dual effect on wax precipitation. Under flash separation conditions, where gas and oil remain in contact, CO₂ acts as a light component at high pressure, inhibiting crystallization. Simultaneously, it extracts light hydrocarbons from the oil phase, which makes the remaining oil heavier and more prone to waxing. The competition between two mechanisms results in a nonmonotonic waxing trend, with a peak in crystallization observed at 30% CO₂ concentration.
- The thermodynamic pathway of gas–oil separation is critical for accurately predicting wax deposition. Initial simulations corresponding to flash separation contradicted field observation at high CO₂ levels. However, the differential separation (degas) process successfully coincided with the field observation, highlighting that the differential separation model is more representative of actual wellbore fluid dynamics.

Based on these findings, practical strategies for mitigating wax deposition can be proposed. Maintaining high pressure and temperature within the wellbore through production optimization can help keep waxes dissolved. Furthermore, managing the CO₂ content in the produced gas, potentially keeping it below 30%, could help prevent the exacerbation of wax precipitation.

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